A Later Test Set Is Not a New Domain: Pretraining Familiarity Survives a Contamination-Free Hold-Out

Mahdi Naser Moghadasi

September 8, 2026

Abstract

Time-series foundation models are evaluated almost exclusively on public archives that predate them, so a strong score cannot be separated from having seen the test set during pretraining. The obvious remedy is a hold-out that postdates the models. We build one: thirteen forecasters — four classical, three trained per dataset, six pretrained — on seven groups drawn from five domains, every observation published after the last model was released, and every dataset rebuildable without an API key. Under this protocol pretrained models win 5 of 7 groups, lose one to a Theta baseline, and on daily exchange rates are indistinguishable from a seasonal naive forecast, along with every other method tested.

We then ask what separates the wins from the losses, and report a negative result: the two intrinsic properties one would reach for — seasonal strength and spectral entropy, measured on the input window — do not account for the pattern, and seasonal strength is if anything negatively associated with the advantage. What does track it is corpus familiarity. Our largest gain (28% lower MASE than the best classical method, on weekly Wikipedia pageviews) falls on Wikipedia pageviews, the domain TimesFM's authors describe as the bulk of its pretraining corpus, at the same granularities and differing only in time window. Within the pretrained family, where every model forecasts identical series so that series difficulty cancels, the TimesFM family outranks the Chronos family by -0.53 on Wikipedia against -0.09 everywhere else (1,500 vs. 754 series, Mann–Whitney $p < 10^{-5}$). We conclude that a temporal hold-out removes memorisation of a window but not familiarity with a domain, that benchmarks therefore need domain hold-outs stated relative to disclosed corpora, and that the practitioner's question is less which model is better than whether their domain is one the model was raised on.

1 Introduction

The claim attached to time-series foundation models is that a single pretrained network forecasts unseen series without being fitted to them, and that it does so better than methods estimated on the target series itself. That claim is usually supported by evaluation on a standard collection of public archives. Those archives are the difficulty. Every widely used benchmark set — the M-competition data, ETT, the Monash repository — predates the models being evaluated and is distributed in the same public form that pretraining corpora are assembled from. A high score is therefore consistent with two very different explanations, generalisation and memorisation, and the published evaluations cannot distinguish them.

The remedy is not a better statistic but a better hold-out. This paper evaluates on observations that did not exist when the models were trained. We assemble seven forecasting groups from five domains, every one of them published continuously by an institutional source, and place the entire test window after the release date of the most recently released model in the comparison. Nothing in the hold-out can have been memorised because none of it had happened yet.

Under that protocol the headline finding of the foundation-model literature survives only in part. Pretrained models are clearly better on some domains and clearly not better on others. The interesting question is what separates them, and our answer is not the one we set out to give: we expected a property of the series and found a property of the corpus. The domain on which pretrained models gain most is the domain one of them was largely pretrained on — the same source and the same granularities, differing only in time window — and the advantage is largest for exactly the model whose corpus it is.

This is not memorisation of the test set; the hold-out makes that impossible. It is the weaker but more durable form of the same problem. A model that has read a decade of Wikipedia pageview series has learned what Wikipedia pageview series do, and January 2026 pageviews still do it. Moving the test window forward controls for the first kind of contamination and leaves the second untouched, which means the field's evaluations cannot be repaired by recency alone.

Contributions.

1. **A contamination-free protocol**, with a hold-out that postdates every evaluated model, over five domains built only from keyless, continuously published sources so that any reader can rebuild the exact panel from the fetchers we release.
2. **Two negative results.** Pretrained models do not beat classical baselines everywhere: we identify one domain where a Theta baseline ranks first and one where nothing beats seasonal naive. And the intrinsic explanation fails — neither seasonal strength nor spectral entropy accounts for where the advantage appears.
3. **Evidence that pretraining familiarity persists past a temporal hold-out**, from a within-family comparison on identical series that no property of those series can explain.
4. **Significance rather than decimal places.** We report multiple-comparison rank intervals and corrected paired tests, and show that most published-style gaps between leading pretrained models fall inside them.
5. **Cost alongside accuracy**, measured on identical hardware in the same runs.

2 Related work

Pretrained forecasters. Chronos [Ansari et al., 2024] tokenises scaled series and trains a language-model architecture over the token sequence; Chronos-Bolt and Chronos-2 are its faster successors. TimesFM [Das et al., 2024] is a decoder-only patched transformer trained on a large corpus dominated by web and search traffic. Moirai [Woo et al., 2024] targets arbitrary frequency and multivariate input via masked patch prediction. These models are trained on differing and only partially disclosed corpora, which is exactly why an evaluation cannot verify by inspection that a given test set was excluded.

Benchmarks and their contamination. The M4 [Makridakis et al., 2020] and M5 [Makridakis et al., 2022] competitions established the evaluation conventions this paper follows: MASE and sMAPE against a seasonal naive scale, and multiple-comparison rank intervals rather than raw mean losses. Their data, along with the Monash archive [Godahewa et al., 2021], is public and static, which makes it ideal for comparability and unusable for testing memorisation. Recent work has begun to raise leakage explicitly; our contribution is to make the hold-out temporal and the collection reproducible rather than to audit corpora post hoc.

Classical baselines. Theta [Assimakopoulos and Nikolopoulos, 2000] won M3 and remains difficult to beat on seasonal series; automatic ARIMA and ETS [Hyndman et al., 2008] are the standard automated alternatives. We use the `statsforecast` implementations so that the baselines are the ones practitioners would actually run.

3 Protocol

3.1 Contamination-free hold-out

The forecast origin is fixed at 1 January 2026 for every group; each model receives the observations preceding it as context and is scored on the horizon that follows. Every evaluated checkpoint — `amazon/chronos-bolt-small` and `-base`, `amazon/chronos-2`, `google/timesfm-2.5-200m-pytorch`, `google/timesfm-3.0-pytorch` and `Salesforce/moirai-2.0-R-small` — was published before that date, so no test observation existed when any of them was trained.

We deliberately do not tabulate pretraining cutoffs. Several of these models disclose their corpora only in summary, and a table of dates assembled from release notes would give the protocol a precision it does not have. The argument does not need it: an observation recorded in 2026 cannot appear in a checkpoint published in 2025, whatever that checkpoint's internal cutoff was. What we can pin exactly is the software, and `requirements-lock.txt` records every library version used to produce these numbers.

3.2 Domains

All five domains are published continuously by an institution, are retrievable without registration or an API key, and are not redistributions of an existing benchmark. Table 1 lists the resulting groups.

Wikipedia pageviews (daily, weekly, monthly). Human attention: strongly periodic, heavy-tailed, punctuated by exogenous spikes.

Weather (hourly, ERA5 reanalysis via Open-Meteo). Smooth, dominated by a clean daily cycle.

Air quality (hourly, CAMS via Open-Meteo). Same sampling rate and daily cycle as weather, but spiky and heavy-tailed. Included specifically to separate "good at hourly data" from "good at smooth data".

Electricity (hourly, Danish grid settlement). The domain most often claimed in the foundation-model literature, but evaluated here without the contamination that the ETT datasets carry.

Exchange rates (daily, ECB reference rates). The adversarial case: the standing result in forecasting is that nothing reliably beats a naive forecast, so this domain tests whether a benchmark can detect the absence of an effect.

Series selection is specified before any result is seen — the full city list, the full currency list, the publisher's own column set — and nothing is added or removed afterwards. One mechanical exclusion applies: a series taking fewer than three distinct values across its entire input window is dropped, because MASE divides by an in-sample seasonal-naive error that is then zero and the metric is undefined. Exactly one series in the study meets this condition (a hydro-generation column for a Danish price area with no hydro, reported as zero throughout), and the rule is stated in these terms rather than as a threshold because a relative one cannot catch it: its scale is negligible in absolute terms and large relative to itself.

3.3 Metrics and significance

We report MASE and sMAPE as defined by the M4 organisers, and weighted quantile loss and 80% interval coverage for the probabilistic models. Because a table of mean losses invites ranking by the third decimal place, our primary comparison is by rank: series are ranked across models, and each model receives an average rank with a Nemenyi interval derived from the Friedman statistic, following the multiple-comparisons-with-the-best presentation used to report M5. We additionally test each model against the per-group winner with a Wilcoxon signed-rank test, Holm-corrected across comparisons. We use signed-rank rather than a paired t-test because MASE across a panel is strongly right-skewed, and we avoid the label "Diebold–Mariano" because that test compares errors over time within one series, which is not the comparison a panel benchmark makes.

The gap between a mean and a rank is not academic here, and we met it by accident. Before the exclusion above was in place, the near-constant Danish hydro series gave the two neural models mean MASE values of order 10^6 on the electricity group — one series in forty-seven moving the group mean by six orders of magnitude, while every model's typical behaviour on that group was unremarkable. The rank-based conclusion for that group was identical with and without the series. A benchmark reported as a mean is one degenerate series away from a headline; reported as ranks it is not.

4 Results

4.1 Accuracy

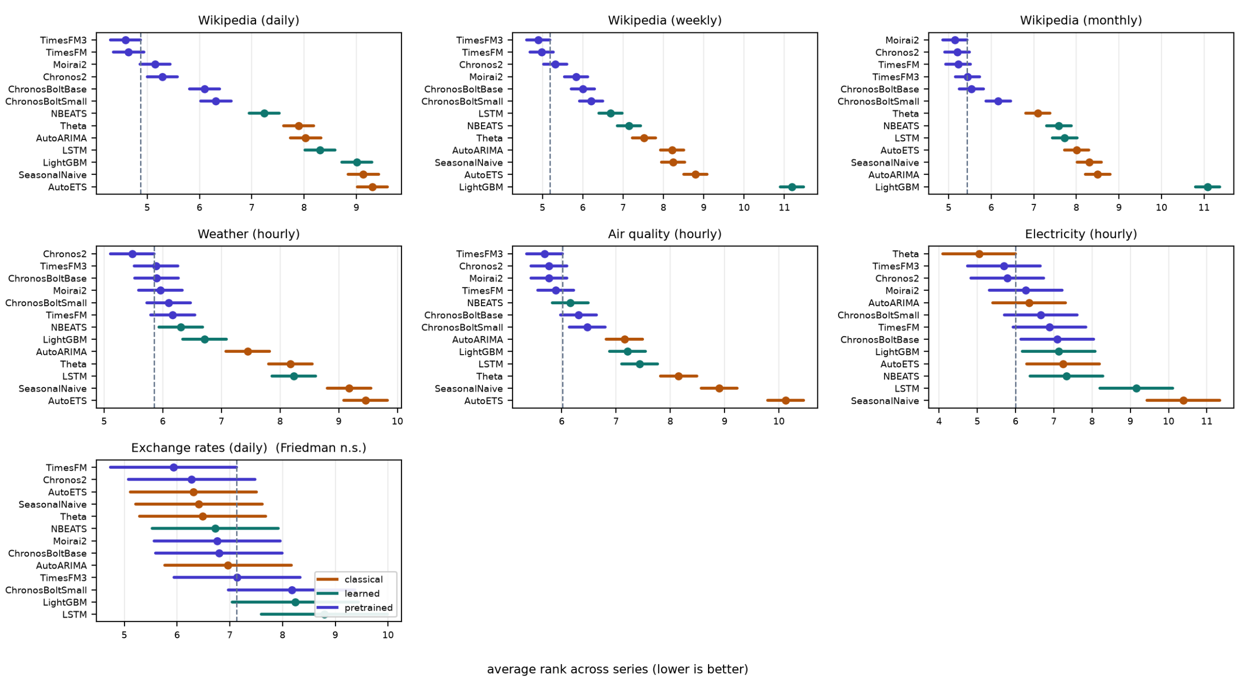

Figure 1: Average rank with Nemenyi intervals, per group. Any model whose interval crosses the dashed line is statistically indistinguishable from the best model in that panel. Exchange rates, on the bottom row, is the panel where every interval overlaps every other and the Friedman test does not reject.

Model	Wikipedia (daily)	Wikipedia (weekly)	Wikipedia (monthly)	Weather (hourly)	Air quality (hourly)	Electricity (hourly)	Exchange rates (daily)
Classical							
SeasonalNaive	1.080	1.642	0.822	1.359	1.142	1.443	2.405
Theta	0.961	1.499	0.746	1.280	1.120	**1.045**	2.448
AutoETS	1.149	1.817	0.941	1.558	1.514	1.255	2.468
AutoARIMA	0.928	1.505	0.853	1.191	1.024	1.183	2.502
Trained per dataset							
LightGBM	1.162	3.772	1.545	1.118	0.983	1.260	3.227
LSTM	0.915	1.202	0.786	1.269	0.988	1.376	2.859
NBEATS	0.870	1.326	0.819	1.093	0.903	1.227	2.539
Pretrained							
ChronosBoltSmall	0.790	1.132	0.641	1.067	0.923	1.200	2.614
ChronosBoltBase	0.774	1.143	0.635	1.055	0.915	1.279	2.501
Chronos2	0.734	1.106	**0.620**	**1.043**	**0.895**	1.155	**2.390**
TimesFM	0.695	1.089	0.621	1.073	0.903	1.201	2.400
TimesFM3	**0.689**	**1.073**	0.623	1.069	0.901	1.157	2.418
Moirai2	0.748	1.135	0.629	1.067	0.906	1.185	2.517

Table 1: Mean MASE by group; lower is better, best per column in bold.

Model	Wikipedia (daily)	Wikipedia (weekly)	Wikipedia (monthly)	Weather (hourly)	Air quality (hourly)	Electricity (hourly)	Exchange rates (daily)
Chronos2	5.29	$\underline{5.31}$	$\underline{5.20}$	$\underline{5.48}$	$\underline{5.76}$	$\underline{5.78}$	$\underline{6.28}$
TimesFM3	$\underline{4.58}$	$\underline{4.89}$	$\underline{5.44}$	$\underline{5.88}$	$\underline{5.68}$	$\underline{5.70}$	$\underline{7.14}$
TimesFM	$\underline{4.64}$	$\underline{4.98}$	$\underline{5.22}$	$\underline{6.17}$	$\underline{5.89}$	$\underline{6.89}$	$\underline{5.93}$
Moirai2	$\underline{5.15}$	5.83	$\underline{5.15}$	$\underline{5.96}$	$\underline{5.76}$	$\underline{6.26}$	$\underline{6.76}$
ChronosBoltBase	6.10	6.00	$\underline{5.53}$	$\underline{5.89}$	$\underline{6.31}$	7.09	$\underline{6.79}$
ChronosBoltSmall	6.31	6.21	6.16	$\underline{6.10}$	6.47	$\underline{6.65}$	$\underline{8.17}$
NBEATS	7.24	7.15	7.59	6.31	$\underline{6.16}$	7.33	$\underline{6.72}$
Theta	7.90	7.51	7.09	8.18	8.15	$\underline{5.04}$	$\underline{6.48}$
AutoARIMA	8.03	8.21	8.50	7.45	7.15	$\underline{6.35}$	$\underline{6.97}$
LSTM	8.31	6.69	7.73	8.24	7.44	9.15	8.79
AutoETS	9.31	8.79	8.01	9.46	10.13	7.24	$\underline{6.31}$
SeasonalNaive	9.13	8.24	8.30	9.18	8.90	10.39	$\underline{6.41}$
LightGBM	9.01	11.19	11.08	6.71	7.21	7.13	$\underline{8.24}$

Table 2: Average rank across series (lower is better). Underlined entries are statistically indistinguishable from the best model in that column under the Nemenyi interval.

Table 2 is the paper's central result, and it does not read like the tables in the papers introducing these models. A pretrained model holds the top rank in 6 of the 7 groups, but that count overstates the case and we do not use it: in 1 of those the Friedman test does not reject, so the column has a winner in the arithmetic sense only. Counting a rank-first under a null result as a win is precisely the overclaim this paper exists to argue against. The defensible statement is that pretrained models win 5 groups, lose one, and are indistinguishable from everything else — seasonal naive included — on the last.

The two exceptions are the informative ones. On electricity Theta ranks first and the leading pretrained models are merely indistinguishable from it. On exchange rates the tied set contains every method tested; the benchmark detects no difference between any of the thirteen, which is the right answer for that domain rather than a failure of the design.

4.2 The intrinsic explanation, and why we abandoned it

Ordering the groups by the gap between the best pretrained model and the best classical one gives a gradient: largest on Wikipedia traffic, smaller on weather and air quality, negative on electricity, null on exchange rates. The natural hypothesis is that some property of the series explains it, and the natural candidates are periodicity and noise — pretraining should help most where a repeating structure exists but is hard to estimate from a few hundred observations alone.

We tested that directly. For each group we measured, on the input window only, the median seasonal strength (from an STL decomposition at the group's seasonal period) and the median spectral entropy of the differenced series, both as defined in the feature literature used to characterise the M4 panel (Table 3). Neither explains the pattern. Seasonal strength is if anything *negatively* associated with the advantage: Wikipedia series have almost no measurable seasonality at the tested

Group	Seasonal strength	Spectral entropy	Pretrained gain
Wikipedia (daily)	0.10	0.93	+25.8%
Wikipedia (weekly)	0.00	0.92	+28.4%
Wikipedia (monthly)	0.06	0.87	+16.9%
Weather (hourly)	0.77	0.69	+12.5%
Air quality (hourly)	0.34	0.84	+12.6%
Electricity (hourly)	0.47	0.83	-10.6%
Exchange rates (daily)	0.00	0.93	n.s.

Table 3: Series properties measured on the input window only, against the observed advantage. Neither column orders the last one.

period and show the largest gains, while weather is the most strongly seasonal group in the study and shows a middling one. Spectral entropy fails in the other direction: exchange rates are as high-entropy as Wikipedia traffic and show no advantage at all. With seven groups no correlation here is statistically meaningful in any case, and we report the measurement rather than a story fitted to it.

4.3 Corpus familiarity survives the hold-out

What does line up with the gradient is not a property of the series but of the models' training data. TimesFM's authors describe its pretraining corpus as dominated by Wikipedia pageviews — on the order of 10^{11} time points at hourly, daily, weekly and monthly granularity, running to late 2023. Our three Wikipedia groups are pageviews at daily, weekly and monthly granularity, beginning in 2026. The hold-out is clean in time and identical in kind.

A domain being both the largest corpus component and the largest measured gain is suggestive, but it is one coincidence and admits a benign reading: perhaps Wikipedia traffic is simply a domain where all pretrained models do well. We can separate the two, because the pretrained models forecast the *same series*. Any property of those series — difficulty, seasonality, noise — applies equally to all of them and cancels in a within-family comparison. What does not cancel is what each model read during pretraining.

Ranking the six pretrained models against each other on every series, the TimesFM family sits -0.53 ranks below the Chronos family on Wikipedia and only -0.09 below it on the other four domains (1,500 vs. 754 series; Mann–Whitney $p < 10^{-5}$, rank-biserial -0.13). The model with Wikipedia in its corpus is specifically better at Wikipedia, and only there. The effect is modest in size, and we present it as such — but its direction is not something the series can account for.

We therefore read the gradient as a corpus-overlap effect rather than a capability one, and note that this is a hypothesis our design supports rather than a demonstrated mechanism. The decisive experiment is a benchmark whose domains are stratified by membership in each model's disclosed corpus, which requires disclosure the field does not currently provide. That, rather than a further increase in the size of the pretraining set, seems to us the useful next contribution.

4.4 What this means for evaluation

A temporal hold-out is necessary and not sufficient. It rules out the model having seen these observations; it does not rule out the model having spent its training on this kind of series, and the second effect is large enough to reorder a leaderboard. Reporting a single average across domains hides it entirely, because the average is dominated by whichever domains happen to be in the benchmark, and those are the well-published ones — which are also the ones most likely to be in a pretraining corpus.

4.5 Cost

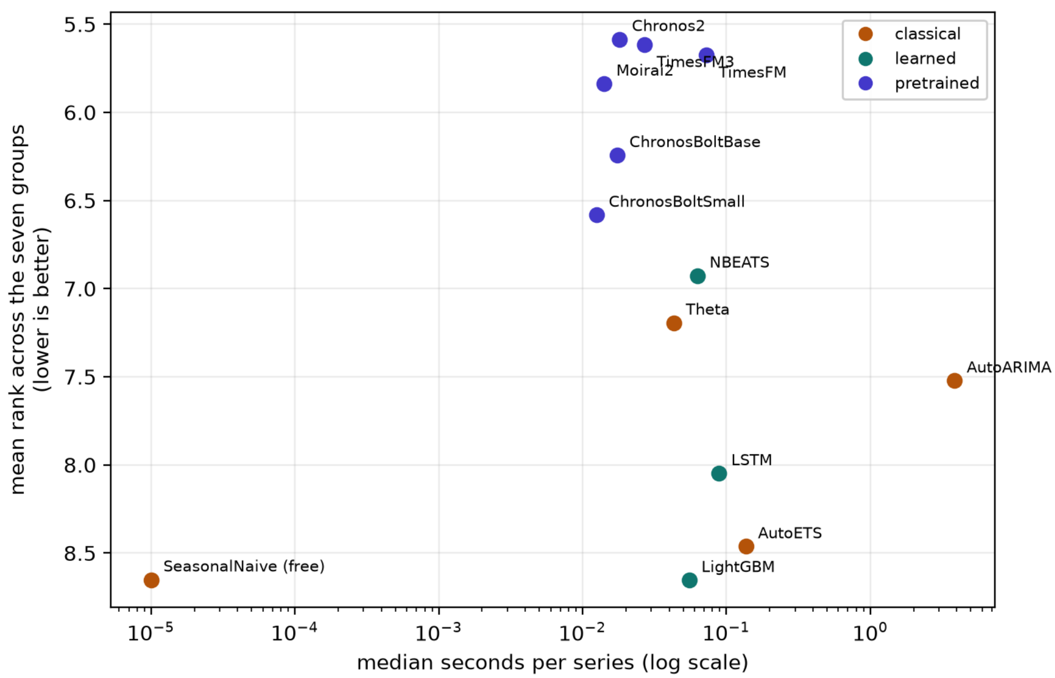

Figure 2: Cost against accuracy, aggregated over the seven groups. The pretrained models occupy a tight cluster at the top left — cheapest and most accurate — while the automatic ARIMA search sits two orders of magnitude to the right of them and ranks worse than a Theta baseline that costs a hundredth as much.

The accuracy differences among the leading pretrained models are small enough to sit inside their rank intervals in most groups. Their costs are not comparable in the same way. An automatic ARIMA search costs a median 54× more per series than a pretrained forward pass, and that factor is not a constant: it runs from 5× on weekly series to 2,689× on weather (hourly), since the order search scales with series length and seasonal period while a forward pass does not. On the 7 groups where it completed, it was beaten on accuracy by the best pretrained model in 7. For a practitioner the comparison that matters is therefore not among the pretrained models, whose differences are mostly not resolvable, but between that family and the automated classical search it would replace — and there the case is strongest exactly where the search is most expensive.

5 Limitations

The corpus-overlap finding is an association, not a demonstrated mechanism. It rests on one corpus whose composition is publicly described and one domain that matches it; a model could be better at Wikipedia traffic for reasons unrelated to having read Wikipedia traffic, and our within-family comparison narrows that possibility without closing it. The effect is also modest — a rank-biserial correlation of -0.13 — and we resist reading a large practical consequence into it. Settling the

Model	Median s/series	Relative to ChronosBoltSmall
SeasonalNaive	0.0000	free
ChronosBoltSmall	0.0125	1.0×
Moirai2	0.0141	1.1×
ChronosBoltBase	0.0174	1.4×
Chronos2	0.0181	1.4×
TimesFM3	0.0269	2.2×
Theta	0.0429	3.4×
LightGBM	0.0550	4.4×
NBEATS	0.0627	5.0×
TimesFM	0.0726	5.8×
LSTM	0.0887	7.1×
AutoETS	0.1362	11×
AutoARIMA	3.8484	308×

Table 4: Median wall-clock seconds per series on identical hardware, including model load.

question needs a benchmark stratified by corpus membership across several models, which in turn needs disclosure that most published corpora do not provide.

Beyond that: the hold-out is one forecast origin per group, where a rolling-origin evaluation would estimate variance across time as well as across series. Two domains contribute fewer than fifty series, so their rank intervals are wide, and the electricity result should be read as "no evidence of an advantage" rather than a demonstrated deficit. Pretraining cutoffs are taken from model documentation and cannot be independently verified; where a cutoff is undisclosed we substitute the release date. And the study covers five domains, more than the standard two but far from the diversity of real forecasting practice — with the caveat, central to our argument, that adding whichever domains are easiest to obtain is likely to add exactly the well-published ones that pretraining corpora already contain.

6 Conclusion

Evaluated on data that postdates them, time-series foundation models are strong on some domains, unnecessary on others, and useless where nothing works. We set out to explain that spread with a property of the series and could not: neither periodicity nor entropy orders it. What orders it is which domains the models were raised on. The largest advantage in our study falls on the domain that constitutes the bulk of one model's pretraining corpus, at the same granularities, and it is that model's family which holds the advantage there and nowhere else.

The practical consequence for benchmark design is that moving the test window forward is not enough. It is necessary — without it, nothing can be concluded at all — but it controls only for having seen these observations, not for having spent a hundred billion time points learning what this kind of observation looks like. A benchmark that wants to measure generalisation has to hold out domains, not just dates, and it can only do so if pretraining corpora are disclosed well enough to say which domains those are. The practitioner's version of the same point is shorter: before asking which model is best, ask whether your data looks like the data it was raised on.

We release the fetchers, the harness and the per-series losses so that both the negative result and the corpus effect can be tested on domains we did not consider.

Reproducibility

Every dataset is rebuilt by a script in `data/sources/` requiring no credentials. `scripts/run_holdout.sh` reproduces the runs, `scripts/significance.py` the tests, and `scripts/make_tables.py` every table and every number quoted above. Per-series losses are released alongside the aggregates so that any alternative test can be applied without re-running the models.

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